

SHRISH KUMAR SINGHAL

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Authorized to work in the U.S. (STEM OPT) | Eligible for the UK High Potential Individual visa

SUMMARY

Builds models, validates them against alternatives, and runs them in production. Implemented option valuation three independent ways as mutual cross checks, and a Treasury curve bootstrap producing the discount and forward rates a swap is valued off. MS in Computer Science from NYU, B.Tech from IIT Guwahati, FRM Part I passed in the top quartile of all four domains.

EDUCATION AND CERTIFICATION

Financial Risk Manager (FRM), Part I

Passed August 2026

Global Association of Risk Professionals | Digital Badge

- Top quartile (76th to 100th percentile) in **all four** exam domains: Foundations of Risk Management; **Quantitative Analysis; Financial Markets and Products; Valuation and Risk Models.**

New York University, Tandon School of Engineering

Sept 2023 - Jan 2025

Master of Science in Computer Science | GPA: 3.83 / 4.0

New York, NY

- Coursework: Intro to Quantitative Finance, Machine Learning, Deep Learning, Artificial Intelligence, Big Data, Data Engineering

Indian Institute of Technology (IIT) Guwahati

July 2019 - May 2023

B.Tech. Electronics and Communication Engineering; Minor in Computer Science

Guwahati, India

- Admitted via JEE Advanced at rank **1,395 of 1.5 million** candidates, top 0.1 percent nationally.

EXPERIENCE

AI Engineer II (Previously Data Scientist & AI Engineer)

June 2025 - August 2026

Boson Motors

San Jose, CA

- Owned a **daily production process end to end** over **50,000+** records per hour, from data collection through generation to quality control, with completeness and staleness tests on each feed, distributional checks on output, and cross-source reconciliation that removed **100%** of a recurring manual verification step.
- **Investigated variance and anomalies to root cause** rather than correcting the symptom, tracing suspect figures back through the transformation chain, escalating when issues were material, and adding controls that prevented the same breaks from recurring.
- **Developed, tested, documented, and monitored models in live use**, with automated retraining, drift detection between expected and realized behavior, and documented validation checks on every rebuild, sustaining above **95%** accuracy in production.
- Selected estimation techniques based on evidence: paired **classical statistical methods** (Isolation Forest, distributional tests) with **machine learning** (LSTM autoencoders) and benchmarked them against each other. Delivered forecasts that predicted failures two weeks ahead and cut unplanned downtime by **34%**, alongside a reinforcement learning framework (Soft Actor-Critic) calibrating 10+ parameters for a **25%** improvement.
- Built **LLM systems**: developed an agent with authenticated, permission-scoped access to internal platform APIs, and deployed a **retrieval-augmented generation** pipeline over technical documentation, with outputs validated against source data rather than trusted implicitly.
- **Communicated to non-technical stakeholders**, translating methods, assumptions, and failure modes into actionable insights for engineering, operations, and management, while building the dashboards and written summaries they relied on.

Quantitative Researcher Intern

June 2024 - Oct 2024

New York University | Supervisor: Prof. Carlos De Oliveira

New York, NY

- Designed and implemented a **statistical arbitrage strategy** across the NASDAQ, S&P 500 and Dow Jones over **24 years** of daily data, writing the full research stack in Python: ingestion and validation, feature construction, a custom back testing engine, and daily P&L, exposure and risk reporting. **12% CAGR in backtest, net of 0.5% assumed transaction costs.**

- Researched **US Treasury rates as a macro driver**, forecasting the 10 year yield over a 100 day forward horizon and integrating it through SQL and pandas to condition positioning on the rate environment, then **isolated its marginal contribution** at roughly 2 percentage points of annual return rather than assuming it helped.
- **Implemented an alternative benchmark and reported the comparison honestly**, running a classical statistical specification (SARIMA) against a machine learning one (LSTM) on identical data and stating which won.
- Evaluated across **18 consecutive quarters** chosen to include the 2020 drawdown and the 2022 rate hiking cycle, reported the worst quarter at **-9.9%** alongside the headline figure, and built a risk overlay that adjusts exposure when realised risk breaches a calibrated tolerance.

PROJECTS

- **US Treasury Curve, Carry and Roll-Down** (Python, NumPy, SciPy, pandas) | **2026**. Bootstraps a zero curve from the FRED constant-maturity par series, then computes present values, duration, DV01, and carry and roll-down on repo-financed Treasuries, extending to DV01-neutral spread and fly economics. **Sixteen assertion-based checks** on the numerics, with known limitations documented rather than hidden.
- **Option Pricing and Cross Validation** (Python, NumPy, pandas) | **2024**. Values European and American options three independent ways, Black-Scholes closed form, Cox-Ross-Rubinstein binomial lattice and Monte Carlo, using each as an accuracy check on the others.
- **Model Comparison and Back Testing Framework** (Python, pandas, NumPy) | **2024**. Abstract interface where a model is specified once and the engine handles ingestion, screening for stale instruments, exposure sizing and daily accounting, so competing specifications run on identical data.

TECHNICAL SKILLS

Quantitative: Option pricing (Black-Scholes, binomial lattice, Monte Carlo), par-to-zero curve bootstrapping and discounting, forward pricing and cash-and-carry, duration, convexity and DV01, carry and roll-down, P&L and exposure attribution, back testing

Statistics and machine learning: Probability theory, econometrics and time series (ARIMA, SARIMA, LSTM), regression, classification, clustering, anomaly detection (autoencoders, Isolation Forest), deep learning (CNN, attention), reinforcement learning, transformers, LoRA fine tuning, retrieval augmented generation, model evaluation and drift monitoring

Programming: Python (pandas, NumPy, SciPy, statsmodels, scikit-learn, PyTorch, Polars), SQL, R, MATLAB, LaTeX; modular design, version control and code review with Git, Docker, Linux

Data and reporting: ETL design, data quality validation and reconciliation, distributed processing (Spark, PySpark, Polars), relational and NoSQL stores, vector search; Tableau, Plotly, Streamlit, Grafana, Excel

PUBLICATIONS AND AWARDS

- S. K. Singhal, C. De Oliveira. “**An LSTM Case Study for Trading American Liquid Indices.**” *Working paper, New York University*, 2024.
- S. K. Singhal et al. “**Segmentation of Bone Tissue from CT Images.**” *Springer Lecture Notes in Networks and Systems*, vol. 586, pp. 233 to 242, 2023. And “**Security Analysis of Aadhaar Authentication Process and Way Forward**”, *IEEE ICAC3N* 2021, sole author.
- **Academic:** JEE Advanced rank 1,395 of 1.5M candidates (top 0.1%); KVPY Science Fellow 2018 and 2019 (top 0.5% of 50,000); NTSE Scholar; Mitacs Globalink Research Fellow, Canada 2022. Top 10 finalist of 2,700+ teams, National Aadhaar Hackathon 2021.